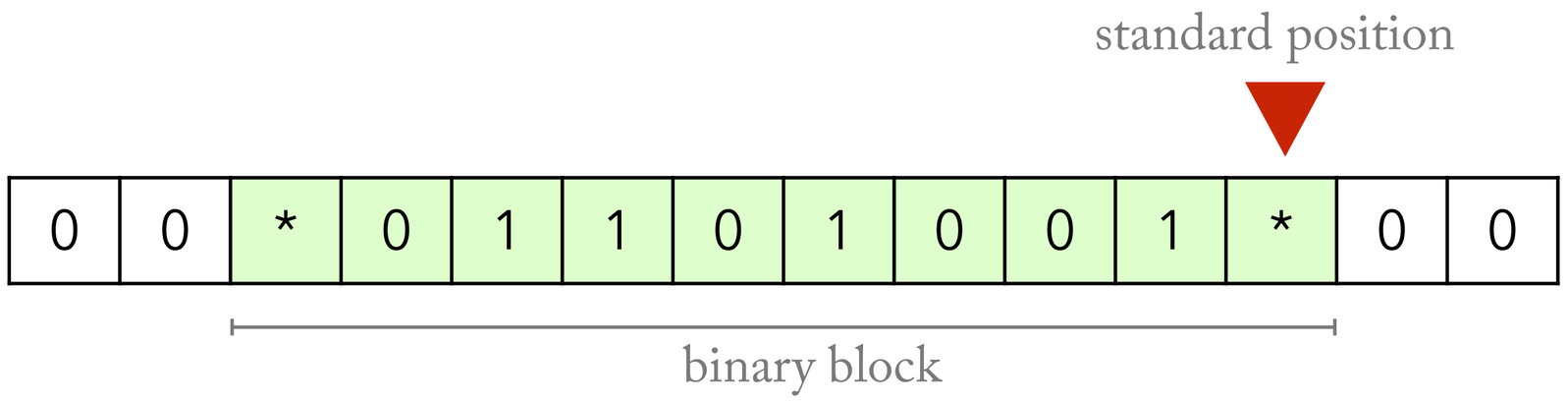
31. Binary Turing Machines

Set-Up

- We now allow three symbols: 1, 0, *
- Turing Machines can read and write all three symbols, so up to *three* arrows can leave each state.
- A **binary block** is a sequence of one or more 1's and 0's, flanked on both sides by *'s:



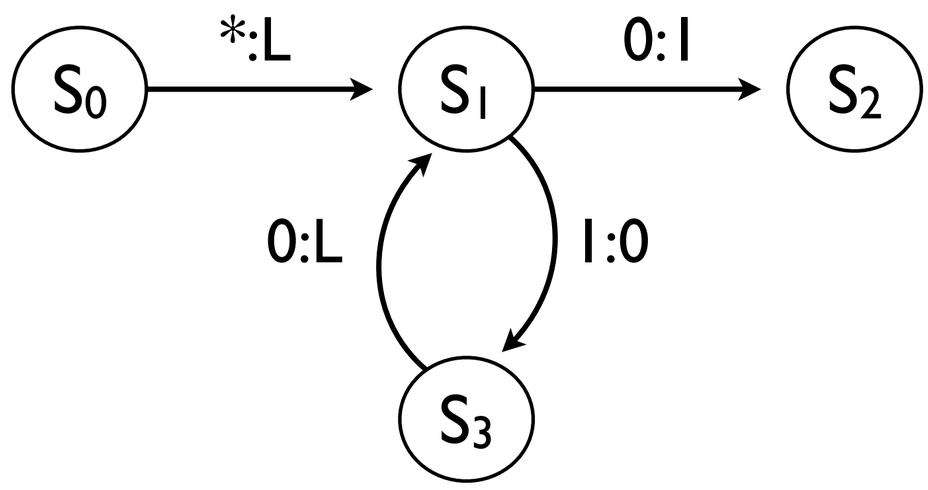
- **Standard position** is defined as the right-most * of the right-most block

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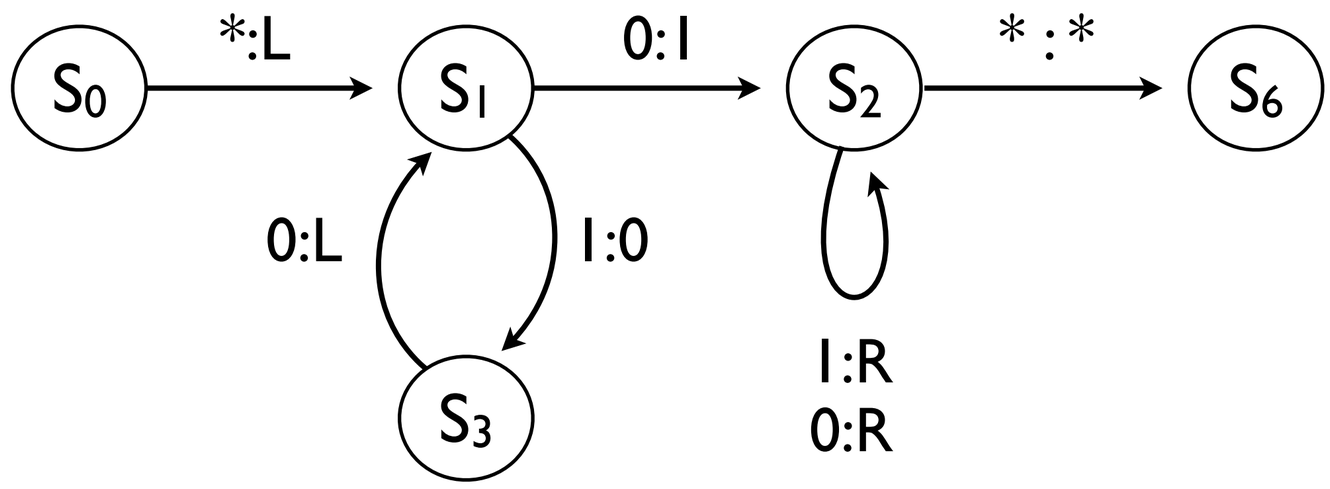
Binary +1

This machine computes the core of Binary +1.

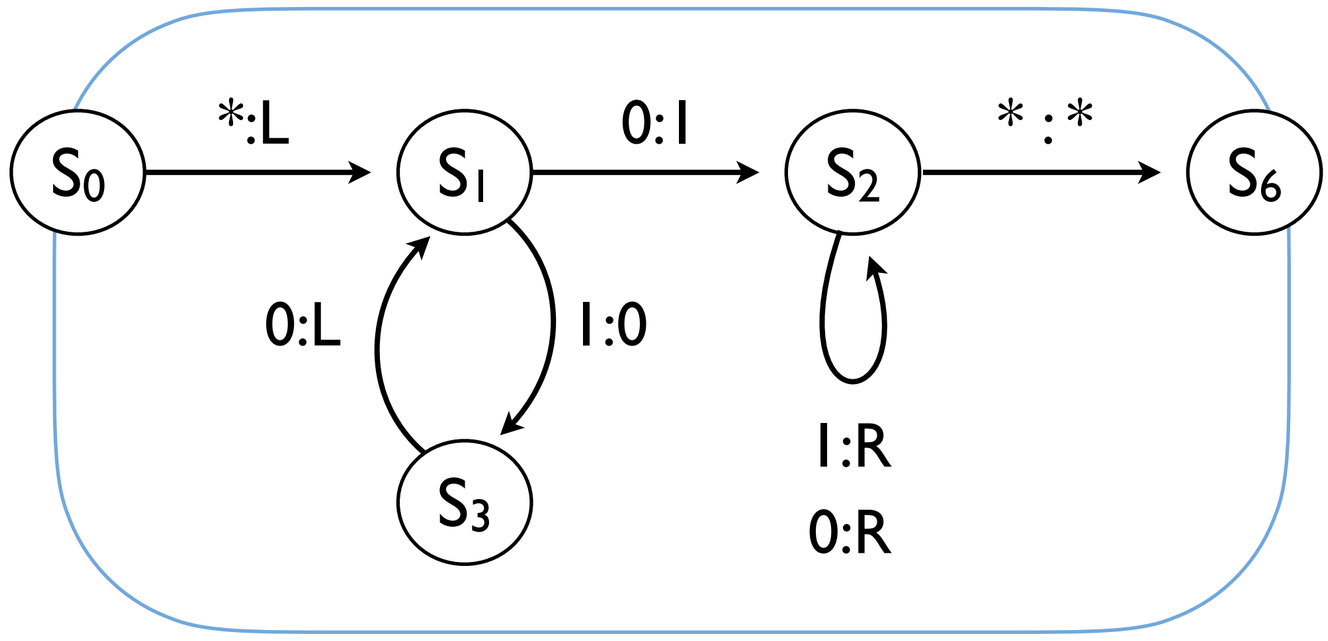
But it doesn't deal with the case where the input's are all 1's, and the left-most * must be overwritten with a 1.



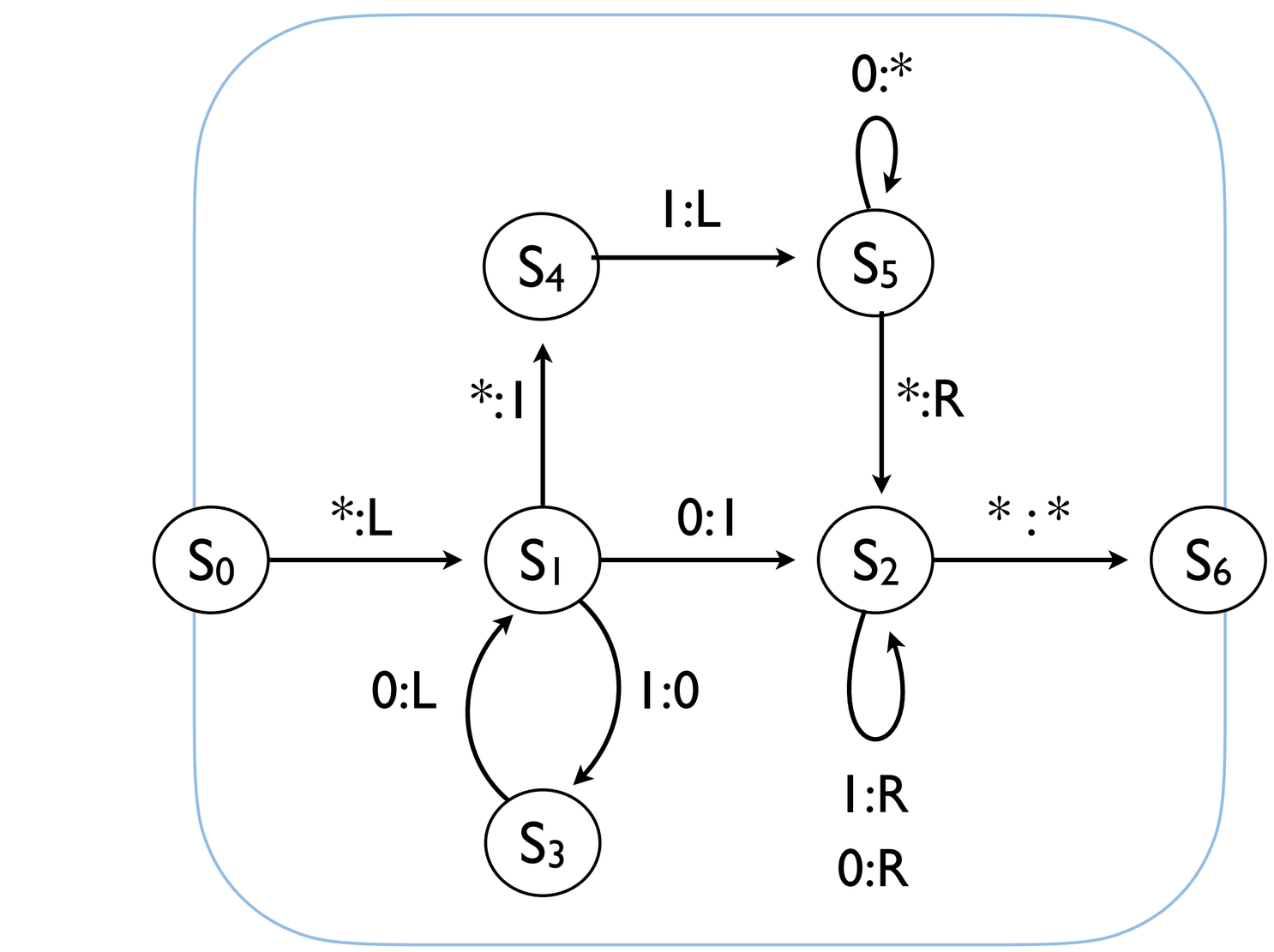
To facilitate boxing, let it return to standard position. Let's also add a terminal node.



This machine can be boxed:



And composed:



When dealing with unbounded binary problems, there is a wrinkle. Suppose the whole binary block is filled with 1's. To compute successor on this numeral, we must overwrite the left-most * with a 1, it then rewrites the * one cell to the left.